

R Programming

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Tutorial Two

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1 Creating Datasets

The first step in any data analysis is the creation of a dataset containing the information to be studied, in a format that meets your needs. In R, this task involves the following:

- Selecting a data structure to hold your data
- Entering or importing your data into the data structure

1.1 Data structures

R has a wide variety of objects for holding data, including **vectors**, **matrices**, **arrays**, **data frames**, and **lists**. They differ in terms of the type of data they can hold, how they're created, their structural complexity, and the notation used to identify and access individual elements.

1.2 Data Input

Data comes to us from a variety of sources and in a variety of formats. Our task is to import the data into our statistical tool, analyze the data and report on the results. R provides a wide range of tools for importing data.

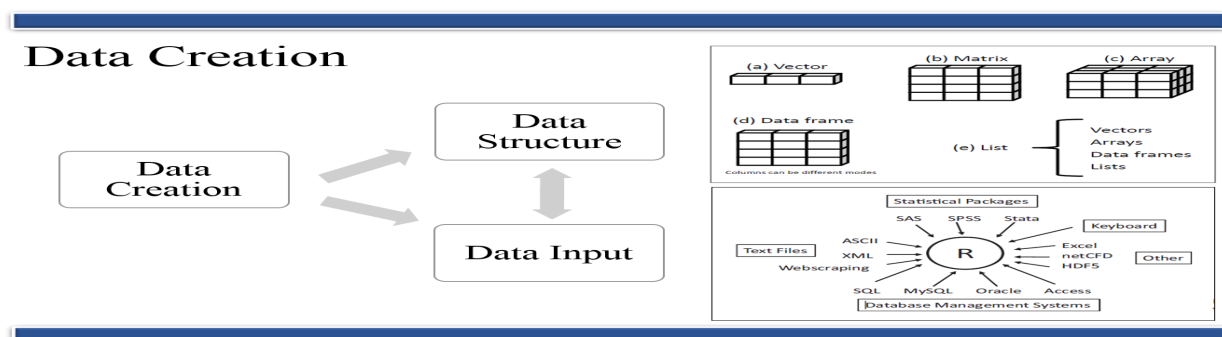


Figure 1: Creating Datasets

2 R Objects

An object is anything that can be assigned to a variable

R objects come in a variety of types. Given an object, the functions **mode** and **class** tell us about its type.

- **mode(object)** has to do with how the object is stored. R has 5 basic modes:
 - **Numeric**
 - **Integer**
 - **Character**
 - **Logical**
 - **Complex**
- **class(object)** gives the class of an object. The main purpose of the class is so that generic functions (like **print** and **plot**) know what to do with it.

Objects could be

- Constants: 2, 13.005, "January"
- Special symbols: NA, TRUE, FALSE, NULL, NaN
- Things already defined in R: seq, c (functions), month.name, letters (character), pi (numeric)
- Data Structures
- Plots
- Functions

3 Data Structures

3.1 Vectors

The vector type is really the heart of R. Vectors are **one-dimensional** arrays that can hold **numeric data, character data, or logical data**. The concatenate function **c()** is used to form the vector

```
1 x <- c(1,2,3) #Numeric
2 y <- c("one", "two", "three") #Character
3 z <- c(TRUE, TRUE, FALSE) #Logical
```

Note that the data in a vector must **only be one type** or mode (numeric, character, or logical). You can refer to elements of a vector using a numeric vector of positions within brackets **x[c(1,3)]** refers to the **1st & 3rd** elements of vector x

3.2 Matrices

A **matrix** is a **two-dimensional array** where each element has the same mode (numeric, character, or logical). Matrices are created with the `matrix` function. The general format is

```
mymatrix <- matrix(vector, nrow=number_of_rows, ncol= number_of_columns, byrow=
logical_value, dimnames=list( char_vector_rownames, char_vector_colnames))
```

```
1 y1<-matrix(1:30,nrow=5,ncol=6) #Way1
2 y2<-1:30
3 dim(y2)<-c(5,6) #Way2
4 y3<-cbind(1:5,6:10,11:15,16:20,21:25,26:30) #Way3
```

You can identify rows, columns, or elements of a matrix by using subscripts and brackets

- `y1[2, ]` #Second row of matrix
- `y1[,3]` #Third column of matrix
- `y1[2,3]` #Element in the second row & third column of matrix
- `y1[3,c(1,5)]` #Elements in third row & 1st & 5th columns of matrix

3.3 Arrays

Arrays are similar to matrices but can have **more than two dimensions**. They're created with an `array` function of the following form:

```
myarray <- array(vector, dimensions, dimnames)
```

```
1 dim1<- c(A1, A2)
2 dim2<- c(B1, B2, B3)
3 dim3<- c(C1, C2, C3, C4, C5)
4 <- array(1:30, c(2,3,5), dimnames=list(dim1,dim2,dim3))
```

Like matrices and vectors, they must be of a single mode. Extracting elements of an array is similar to that of matrices

3.4 Data Frames

The **most common** data structure in R is **data frame**. A data frame is more general than a matrix in that different columns can contain different modes of data (numeric, character, etc.)

A data frame is created with the `data.frame()` function :

```
mydata.frame <- data.frame(col1, col2, col3,...)
```

```

1 id <- c(1,2,3)
2 name <- c("Peter", "Roger", "Hansen")
3 marks<- c(83, 56, 95)
4 marksheet<- data.frame(id,name, marks)

```

- marksheet[,1:2] #Extracting first 2 columns
- marksheet[1:2,] # Extracting first 2 rows
- marksheet[, c("name", "marks")] # Extracting columns using column names
- marksheet\$marks # Extracting a particular column of a data frame

3.5 Lists

Lists are the **most complex** of the R data types. Basically, a list is an **ordered collection of objects** (components). A list may contain a combination of vectors, matrices, data frames, arrays and even other lists.

You create a list using the **list()** function :

```
mylist <- list(name1=object1, name2=object2, ...)
```

```

1 listR<-list( title=First list, x= c(1,4,5), matrix(1:10,2,5), marksheet=marksheet)

```

Note, extraction of elements from list is slightly different:

- listR[[2]] # Extracting second object of list
- listR[[x]] # Extracting object by name
- listR\$x # Alternate way of extracting single object of list
- listR[[2]][3] # Extracting a particular element in an object of the list
- listR[c(1,3)] # Extracting multiple objects